Software Requirements Specification (SRS) – Canteen Pre Order System

Project: Canteen Pre-Order System
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Revision history

Version	Date	Author	Change summary	Approval
1.0	11-10-2026	Student Team	SRS for Canteen Pre-Order System	Pending

Approvals

Role	Name	Signature / Email	Date
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1. Introduction

1.1 Purpose

This document is a Software Requirements Specification (SRS) for the Canteen Pre-Order System. It defines functional and non-functional requirements, interfaces, security requirements, and verification criteria intended for students, instructors, developers, testers, and canteen staff to use as a reference.

Covers student registration and authentication, menu browsing, cart management, food pre-ordering, payment-status handling, order tracking and history, staff menu management, order processing, and basic operational reporting. It excludes physical food preparation, kitchen operations, and internal implementation of external payment providers.

Students, Canteen Staff, Developers, QA Engineers, System Integrators, Maintenance Teams, and Assessment Evaluators.

List of acronyms: SRS, SAD, STP, UI, API, DB, RBAC, RTM, UAT.

2. Overall description

The Canteen Pre-Order System is a web-based client-server application through which students browse a digital menu, add food items to a cart, place pre-orders, make or record payment status, and track orders. Canteen staff use the system to manage menu items, availability, and incoming orders.

- Student registration and login
- Menu browsing, search, and filtering
- Cart management
- Food pre-order placement
- Payment-status handling
- Order confirmation and tracking
- Order history and eligible cancellation

- Staff authentication and role-based access
- Menu item management and availability updates
- Incoming order queue management
- Order acceptance/rejection and preparation-status updates
- Basic order/sales reporting

- Student: regular user who expects simple navigation, accurate prices, and clear order status.
- Canteen Staff: authorized operational user who manages menu items and processes incoming orders.
- System Administrator: optional privileged user responsible for authorized configuration and maintenance.

Web browser environment on standard laptops, desktops, tablets, or smartphones with network connectivity. A suitable implementation may use React/JavaScript for the frontend, Node.js/Express for the backend, and MongoDB for persistent storage.

The system depends on network availability, supported web browsers, database availability, and authenticated access. Only authorized staff may perform staff operations. The academic implementation may use a simulated or sandbox payment mechanism rather than a production payment provider.

3. External interface requirements

Primary UI: responsive web interface with clear navigation for menu, cart, checkout, order tracking, and staff order/menu management. Forms shall provide validation messages and status information.

- Student/client device with keyboard, mouse, touch screen, or equivalent input
- No specialized hardware is required for the core system
- Optional canteen display/printer is outside the core implementation

- Authentication API for registration and login
- Menu API for retrieving and managing menu items
- Order API for cart/order creation and tracking
- Payment-status API or sandbox integration
- Database interface for persistent storage
- Optional notification service

- HTTP/HTTPS communication between browser and backend API.
- JSON request/response format.
- HTTPS shall be used in deployed environments for protected communication.
- Authentication/session or token information shall be transmitted securely.

Make sure overall there are at least 15 FRs for the overall project, 5 NFRs, 2 security objectives, and 5 security requirements.

4. System features (detailed)

Each requirement below includes acceptance criteria and a reference test case. IDs follow CPS-F-###.

4.1 Authentication

Description: Authenticate students and staff using valid credentials and establish an authorized session.

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
CPS-F-001	The system shall allow a student to register using required details and reject duplicate or invalid registration data.	Functional	High	Student / Business	AC-CPS-F-001: Valid new details create an account; invalid/duplicate details are rejected. Test: TC-AUTH-01	Requires user database
CPS-F-002	The system shall authenticate registered users using valid credentials and establish an authenticated session/token.	Functional	High	Security / Student	AC-CPS-F-002: Valid login succeeds; invalid credentials are rejected. Test: TC-AUTH-02	Authentication service
CPS-F-003	The system shall display available menu items with name, category, description, price, and availability.	Functional	High	Student / Canteen	AC-CPS-F-003: Available menu data is displayed correctly. Test: TC-MENU-01	Menu database
CPS-F-004	The system shall allow students to search or filter menu items by name or category.	Functional	Medium	Student	AC-CPS-F-004: Matching menu items are displayed for a valid search/filter. Test: TC-MENU-02	Depends on menu data

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
CPS-F-005	The system shall allow students to add menu items to a cart, change quantities, and remove items.	Functional	High	Student	AC-CPS-F-005: Cart reflects all valid add/update/remove operations. Test: TC-CART-01	Requires authenticated student
CPS-F-006	The system shall validate item availability and quantity before checkout.	Functional	High	Business / Canteen	AC-CPS-F-006: Unavailable/invalid quantities are rejected. Test: TC-CART-02	Server-side validation required
CPS-F-007	The system shall create an order containing selected items, quantities, calculated total, student details, and order time.	Functional	High	Student / Business	AC-CPS-F-007: Valid checkout creates one order with correct details. Test: TC-ORD-01	Requires database
CPS-F-008	The system shall generate a unique order identifier and display confirmation after successful order placement.	Functional	High	Student	AC-CPS-F-008: Unique order ID and confirmation are displayed. Test: TC-ORD-02	Depends on order creation
CPS-F-009	The system shall allow payment to be initiated or payment status to be recorded for an order.	Functional	High	Student / Business	AC-CPS-F-009: Payment status is associated with the correct order. Test: TC-PAY-01	Sandbox/simulation may be used
CPS-F-010	The system shall allow students to view the current status of an active order.	Functional	High	Student	AC-CPS-F-010: Current status is displayed correctly. Test: TC-TRACK-01	Requires order status

4.2 Menu Browsing & Cart Management

Description: Allow students to browse available food items, search or filter the menu, add items to a cart, change quantities, and remove items before checkout.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
CPS-F-011	The system shall allow students to view previous orders and their details.	Functional	Medium	Student	AC-CPS-F-011: Correct order history is displayed for the logged-in student. Test: TC-TRACK-02	Requires order history
CPS-F-012	The system shall allow cancellation only when an order is in a configured cancellable state.	Functional	Medium	Student / Business	AC-CPS-F-012: Eligible order can be cancelled; non-eligible order is rejected. Test: TC-ORD-03	Depends on order status
CPS-F-013	The system shall authenticate canteen staff and restrict staff functions to authorized accounts.	Functional	High	Security / Canteen	AC-CPS-F-013: Staff login succeeds for authorized user and fails otherwise. Test: TC-STAFF-01	RBAC required
CPS-F-014	The system shall allow authorized staff to add, edit, remove, and update availability of menu items.	Functional	High	Canteen Staff	AC-CPS-F-014: Valid staff CRUD/availability changes are persisted. Test: TC-STAFF-02	Staff authorization required

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
CPS-F-015	The system shall allow staff to view incoming orders with order ID, items, quantities, time, payment status, and current status.	Functional	High	Canteen Staff	AC-CPS-F-015: Order queue displays complete order information. Test: TC-STAFF-03	Staff authorization required
CPS-F-016	The system shall allow staff to update an order through valid states such as Accepted, Preparing, Ready, and Completed.	Functional	High	Canteen Staff	AC-CPS-F-016: Valid transitions are accepted and invalid transitions rejected. Test: TC-STAFF-04	State validation required
CPS-F-017	The system shall prevent unavailable menu items from being newly ordered.	Functional	High	Canteen / Business	AC-CPS-F-017: Attempt to order unavailable item is rejected. Test: TC-MENU-03	Availability must be rechecked
CPS-F-018	The system shall calculate order totals on the server from stored item prices and requested quantities.	Functional	High	Business / Security	AC-CPS-F-018: Server total matches stored prices and quantities. Test: TC-ORD-04	Prevents client-side manipulation
CPS-F-019	The system shall display meaningful validation and error messages for invalid input and unsuccessful operations.	Functional	Medium	Student / Staff	AC-CPS-F-019: Appropriate message is shown without exposing internal details. Test: TC-ERR-01	UI and API validation

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
CPS-F-020	The system shall allow authorized staff to view basic order/sales summaries for a selected period.	Functional	Low	Canteen Staff	AC-CPS-F-020: Authorized staff can retrieve a valid summary. Test: TC-STAFF-05	Depends on order records

4.3 Cart Validation & Order Placement

Requirements: The system shall validate cart items and quantities, prevent unavailable items from being ordered, calculate the order total, and create the order with a unique identifier. IDs: CPS-F-006..CPS-F-008.

4.4 Payment & Order Tracking

Requirements: Support payment initiation or payment-status recording, display order confirmation, allow order tracking and history, and support cancellation when permitted. IDs: CPS-F-009..CPS-F-012.

4.5 Staff Operations & Order Management

Requirements: Provide staff authentication, menu management, order queue processing, order status updates, unavailable-item control, server-side total validation, meaningful errors, and basic reporting. IDs: CPS-F-013..CPS-F-020.

5. Non-functional requirements (detailed)

NFRs below are measurable and tied to test plans. IDs CPS-NF-###.

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
CPS-NF-001	90% of menu and order API requests should respond within 3 seconds under normal academic/demo load.	Performance	High	90th percentile ≤ 3s in representative test. Test: TC-PERF-01
CPS-NF-002	The application should be available during scheduled operating/demo hours except planned maintenance.	Reliability	High	Availability record meets agreed operating target. Test: TC-REL-01
CPS-NF-003	The interface should provide clear navigation, readable status information, and understandable validation messages.	Usability	Medium	Representative users complete core flow without critical usability issue. Test: TC-UX-01

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
CPS-NF-004	The system should maintain timestamped application and order-processing logs sufficient for troubleshooting, security review, and operational auditing.	Audit/Data Retention	High	Required application events can be located and reviewed. Test: TC-OPS-01
CPS-NF-005	The web interface should support common accessibility practices, including keyboard navigation, readable text, labels, and suitable contrast where applicable.	Usability/Accessibility	Medium	Accessibility checks pass for the supported core workflows. Test: TC-UX-01
CPS-NF-009	The design should support growth in users, menu items, and orders without redesigning core modules.	Scalability	Medium	Load/data growth test shows acceptable behaviour. Test: TC-PERF-02
CPS-NF-010	Frontend, backend, database, and authentication responsibilities shall be separated into maintainable modules.	Maintainability	Medium	Design/code review confirms separation. Test: TC-MAINT-01
CPS-NF-006	The system should not create duplicate orders when a request is retried after a temporary network failure.	Reliability	Repeated request does not create duplicate order. Test: TC-REL-02	
CPS-NF-007	The web application should work on current supported versions of Chrome, Edge, and Firefox.	Compatibility	Core workflows pass on supported browsers. Test: TC-COMP-01	
CPS-NF-008	Stored order items, quantities, prices, and totals shall remain internally consistent after an order is placed.	Data Integrity	Stored values match the confirmed order. Test: TC-DATA-01	

5.1. Security

5.1.1 Security Objectives

Security objectives:
1. Protect user accounts and order information from unauthorized access.
2. Ensure only authorized staff can perform menu and order-management operations.
3. Maintain the integrity and confidentiality of order, payment-status, and user data.

5.1.2 Security Requirements

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref
CPS-SR-001	Passwords shall be stored using secure one-way password hashing and never in plaintext.	Security	High	TC-SEC-01
CPS-SR-002	Protected API endpoints shall verify authentication before allowing protected operations.	Security	High	TC-SEC-02

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref
CPS-SR-003	Staff-only endpoints shall enforce role-based authorization.	Security	High	TC-SEC-03
CPS-SR-004	The server shall validate prices, quantities, item IDs, and order totals rather than trusting client values.	Security/Data Integrity	High	TC-SEC-04
CPS-SR-005	Sensitive credentials and authentication tokens shall not be written to application logs.	Security	High	TC-SEC-05
CPS-SR-006	HTTPS shall be used in deployed environments for protected communication.	Security	Medium	TC-SEC-06

6. Quality attributes & Acceptance tests

- Exit criteria for acceptance: All high-priority functional requirements implemented and verified, no critical NFR or security failures, and RTM shows all planned test cases executed with required results.

- Acceptance test suites: Authentication, Menu, Cart, Order Placement, Payment Status, Order Tracking, Staff Operations, Performance, Security, Compatibility, and Usability tests.

7. System models and diagrams

7.1 UML Use-Case diagram

At least 2 UML use case diagrams to be created: one for Student functions and one for Canteen Staff functions.

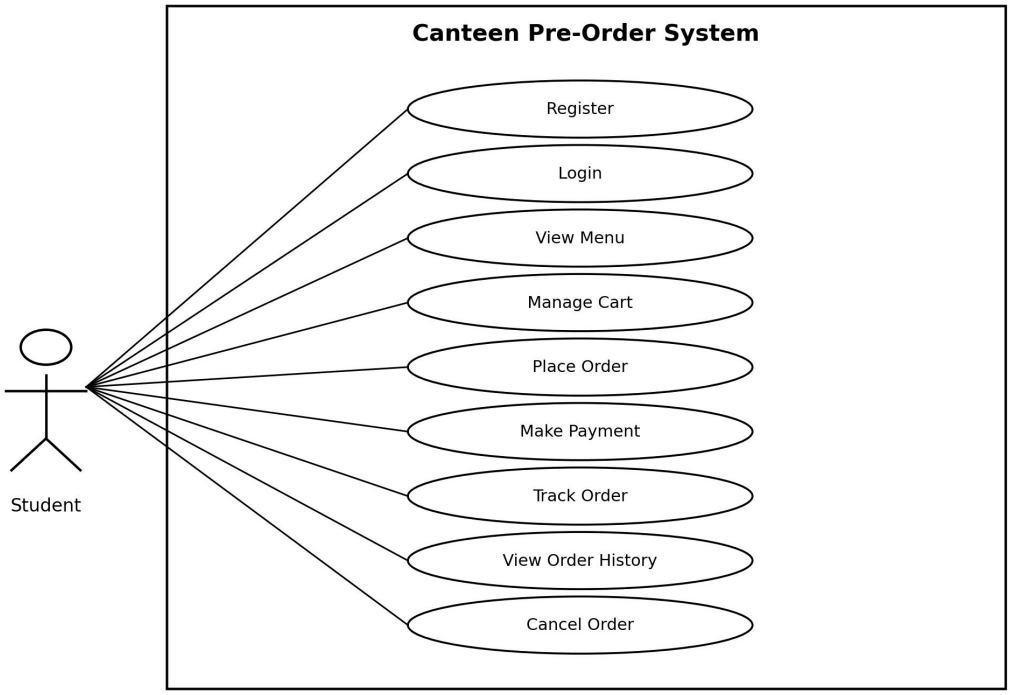


Figure 7.1: UML Use-Case Diagram — Student

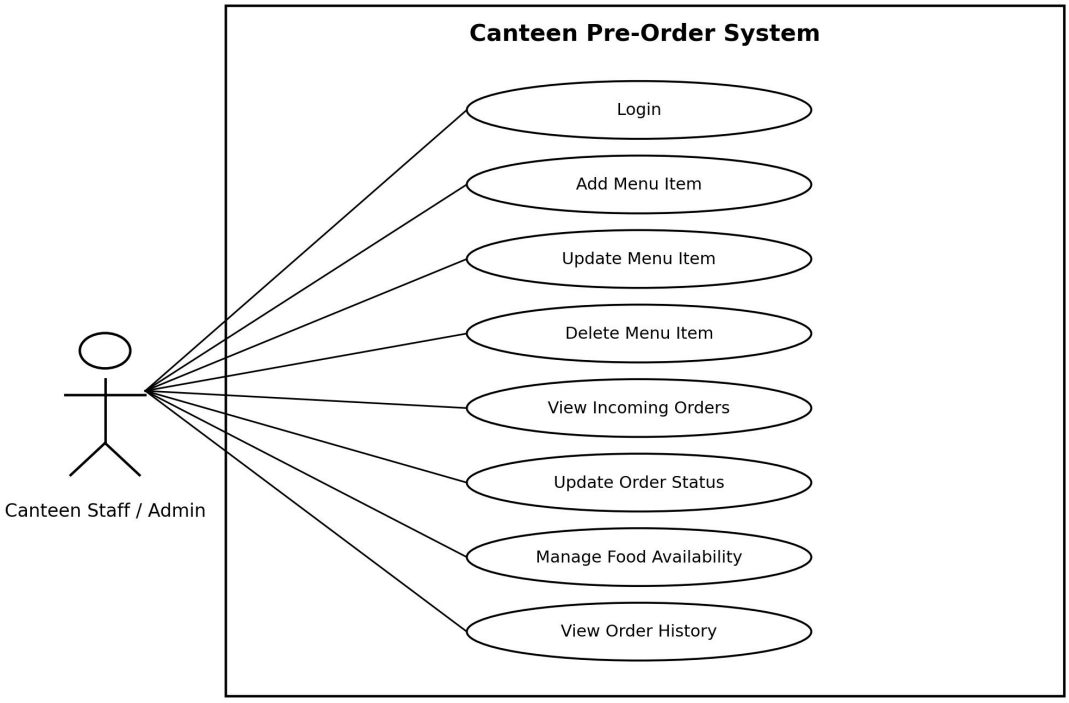


Figure 7.2: UML Use-Case Diagram — Canteen Staff / Admin

8. Requirements Traceability Matrix (RTM)

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
CPS-F-001	Register account	4.1 / DS-Auth-01	AuthModule	TC-AUTH-01	N	
CPS-F-002	Authenticate user	4.1 / DS-Auth-02	AuthModule	TC-AUTH-02	N	
CPS-NF-001	Response time	5 / DS-Perf-01	WebUI / API	TC-PERF-01	N	
CPS-F-003	Display menu	4.2 / DS-Menu-01	MenuModule	TC-MENU-01	N	
CPS-F-005	Manage cart	4.2 / DS-Cart-01	CartModule	TC-CART-01	N	
CPS-F-007	Place order	4.3 / DS-Order-01	OrderModule	TC-ORD-01	N	
CPS-F-009	Payment status	4.4 / DS-Payment-01	PaymentModule	TC-PAY-01	N	
CPS-F-010	Track order	4.4 / DS-Track-01	TrackingModule	TC-TRACK-01	N	
CPS-F-014	Manage menu	4.5 / DS-MenuAdmin-01	StaffModule	TC-STAFF-02	N	
CPS-F-016	Update order status	4.5 / DS-OrderAdmin-01	StaffModule	TC-STAFF-04	N	
CPS-NF-008	Order data integrity	5 / DS-Data-01	Order/DBModule	TC-DATA-01	N	
CPS-SR-003	Staff authorization	5.1 / DS-Sec-01	AuthMiddleware	TC-SEC-03	N	
CPS-SR-004	Server-side validation	5.1 / DS-Sec-02	OrderAPI	TC-SEC-04	N	